

Effect of vitamin C supplements on urinary oxalate and pH in calcium stone-forming patients.

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BACKGROUND: The contribution of ascorbate to urinary oxalate is controversial. The present study aimed to determine whether urinary oxalate and pH may be affected by vitamin C supplementation in calcium stone-forming patients. **METHODS:** Forty-seven adult calcium stone-forming patients received either 1 g (N=23) or 2 g (N=24) of vitamin C supplement for 3 days and 20 healthy subjects received 1 g. A 24-hour urine sample was obtained both before and after vitamin C for calcium, oxalate, magnesium, citrate, sodium, potassium, and creatinine determination. The Tiselius index was used as a calcium oxalate crystallization index. A spot fasting morning urine sample was also obtained to determine the urinary pH before and after vitamin C. **RESULTS:** Fasting urinary pH did not change after 1 g (5.8 ± 0.6 vs. 5.8 ± 0.7) or 2 g vitamin C (5.8 ± 0.8 vs. 5.8 ± 0.7). A significant increase in mean urinary oxalate was observed in calcium stone-forming patients receiving either 1 g (50 ± 16 vs. 31 ± 12 mg/24 hours) or 2 g (48 ± 21 vs. 34 ± 12 mg/24 hours) of vitamin C and in healthy subjects (25 ± 12 vs. 39 ± 13 mg/24 hours). A significant increase in mean Tiselius index was observed in calcium stone-forming patients after 1 g (1.43 ± 0.70 vs. 0.92 ± 0.65) or 2 g vitamin C (1.61 ± 1.05 vs. 0.99 ± 0.55) and in healthy subjects (1.50 ± 0.69 vs. 0.91 ± 0.46). Ancillary analyses of spot urine obtained after vitamin C were performed in 15 control subjects in vessels with or without ethylenediaminetetraacetic acid (EDTA) with no difference in urinary oxalate between them (28 ± 23 vs. 26 ± 21 mg/L), suggesting that the in vitro conversion of ascorbate to oxalate did not occur. **CONCLUSION:** These data suggest that vitamin C supplementation may increase urinary oxalate excretion and the risk of calcium oxalate crystallization in calcium stone-forming patients.